

Pavan Kumar Rao

Senior Game Developer

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PROFESSIONAL SUMMARY

Senior Gameplay Programmer with 8+ years of experience shipping gameplay systems in C++, Unreal Engine 5, and Unity3D. Proven expertise in the Gameplay Ability System (GAS), character movement, AI behaviour, multiplayer architecture, and networked player controls. Strong track record of delivering production-ready features across the full development pipeline, with demonstrated ability to collaborate effectively with cross-functional teams including designers, animators, and artists.

PROFESSIONAL EXPERIENCE

Senior Gameplay Programmer

Dream Game Studios — Pune, India | December 2022 – Present

- Shipped Dream Cricket, an AAA mobile title — architected core character control systems including networked character behaviours, responsive batting and fielding mechanics, and synchronised player controls with client-side prediction and server reconciliation.
- Delivered 5 major gameplay features end-to-end (Async Multiplayer, Player Card Gacha, Team Management, Inventory System, Emote Celebration), owning each from design handoff through production release.
- Enabled seamless PvP/PvE player experience across Quick Match, Tournaments, and Private Lobby by designing a unified game mode architecture that maintained consistent performance across distributed environments.
- Accelerated design iteration speed by building real-time parameter tuning tools and gameplay visualisation overlays, directly shortening the feedback loop between design intent and the player-facing result.
- Improved codebase maintainability and team velocity by introducing MVCS-like frontend architecture; mentored junior engineers on code quality practices; established CI/CD build automation via Jenkins.

Game Developer (SE 2)

Moonfrog Labs — Bangalore, India | January 2020 – November 2022

- Led the migration of Ludo Club (live multiplayer game with active player base) from Cocos2d-x to Unity3D — refactored the networked game loop using Finite State Machine architecture, resulting in improved multiplayer performance and long-term code maintainability.
- Shipped full-stack multiplayer features across client (C#) and server (Node.js) — synchronised achievement and rewards systems with robust error handling under live network conditions.
- Reduced live ops incident response time by building developer and admin tools that improved debugging capabilities for networked gameplay issues.
- Maintained production stability for a live multiplayer title through real-time monitoring, managed deployments, and rapid resolution of critical networking and gameplay bugs.

Associate Game Developer

JumpStart Games — Bangalore, India | January 2018 – December 2019

- Contributed gameplay and social features across multiple shipped Unity3D titles including *Neopets: Legends and Letters*.
- Maintained production stability through ongoing live ops support, delivering rapid bug fixes and feature updates under production pressure.
- Ensured reliable multiplayer experiences by contributing to synchronised game state and social systems across distributed clients.

KEY PROJECTS

Unreal First Person Hack-and-Slash — C++ / Unreal Engine 5

- Built a complete GAS-driven combat system with ability management, cooldowns, attribute replication, and animation-driven combat interactions — abilities trigger animation montages and blend between combat states for physically responsive gameplay.
- Achieved reactive enemy AI through custom State Trees governing decision-making, combat behaviours, and coordinated group tactics.
- Leveraged UE5 Blueprints alongside C++ for rapid prototyping and iteration of gameplay and animation logic.

Top-Down Multiplayer Shooter — Unity3D

- Implemented client-side prediction, server reconciliation, and lag compensation over both peer-to-peer and client-server architectures for responsive multiplayer feel across varying network latencies.
- Delivered smooth, consistent character animation playback in networked play through entity interpolation and synchronised movement and combat actions.

Unity AI Module — C# / Unity

- Improved pathfinding performance in large-scale environments by building a custom A* implementation using binary heap / priority queue optimisation.
- Developed a behaviour tree framework with a custom graph editor for visual debugging and real-time AI behaviour tuning.

2D C++ ECS-Based Engine — C++

- Designed and implemented a custom ECS framework with multithreading and a job system for parallel entity processing.
- Built a custom sequential impulse-based physics system for accurate collision resolution and rigid body dynamics.
- Demonstrated strong foundations in linear algebra, 3D math, memory management, and low-level systems programming.

TECHNICAL SKILLS

Engines	Unreal Engine 5 (C++ & Blueprints), Unity3D (C#)
Core Systems	GAS, Character Movement, Enhanced Input, Camera Systems, AI State Trees, Behaviour Trees
Animation	Animation Montages, State Machines, Animation Blueprints, Networked Animation Sync, Entity Interpolation
Multiplayer	Client-server architecture, client-side prediction, server reconciliation, lag compensation
Languages	C++, C#, JavaScript (Node.js), Python, GLSL
Math & Low-Level	Linear Algebra, 3D Math, ECS, Multithreading, Job Systems, Physics Systems
Tools	Git, Perforce, Jenkins (CI/CD), Agile/Scrum

EDUCATION

B.Sc (Hons) in Game Development

Bharathiar University, Coimbatore | 2014 – 2017